

ADVANCED LINEAR ALGEBRA HOMEWORK 6

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P.4.9. If $A \in \mathbf{M}_n$ is to be represented as $A = XY^\top$ for some $X, Y \in \mathbf{M}_{n \times r}$, explain why r cannot be smaller than $\text{rank}(A)$.

Solution. We claim that the columns of A must lie in the column space of X . To see this we write

$$Y^\top = \begin{bmatrix} y_1^\top \\ y_2^\top \\ \vdots \\ y_r^\top \end{bmatrix},$$

where $y_i \in \mathbb{F}^n$ are columns of Y . Then we have that

$$XY^\top = \sum_{k=1}^r x_k y_k^\top.$$

Now choose an arbitrary vector $v \in \mathbb{F}^n$,

$$Av = XY^\top v = X(Y^\top v)$$

with $Y^\top v \in \mathbb{F}^r$. So Av is a linear combination of the r columns of X , which gives us $\text{col}(A) \subseteq \text{col}(X)$. So

$$\text{rank}(A) = \dim(\text{col}(A)) \leq \dim(\text{col}(X)) = \text{rank}(X) \leq r$$

So $\text{rank}(A) \leq r$, and we are done. □

P.4.10. For $X \in \mathbf{M}_{m \times n}$, $\nu(X)$ denote the nullity of X .

- (a) Show that $\nu(X^\top) = \nu(X) + m - n$.
- (b) Let $A \in \mathbf{M}_{m \times k}$ and $B \in \mathbf{M}_{k \times n}$. Show that the rank inequality given by

$$\text{rank}(A) + \text{rank}(B) - k \leq \text{rank}(AB)$$

is equivalent to

$$(1) \quad \nu(AB) \leq \nu(A) + \nu(B).$$

(c) If $A, B \in \mathbf{M}_n$, show that the rank inequality given by

$$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

is equivalent to

$$(2) \quad \max\{\nu(A), \nu(B)\} \leq \nu(AB)$$

The inequalities (1) and (2) are known as *Sylvester's law of nullity*.

(d) Consider $A = \begin{bmatrix} 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$. Show that the inequality (2) need not be valid if the matrices are not square.

Proof. (a) By the rank-nullity theorem

$$\begin{aligned} \text{rank}(X) + \nu(X) &= n \\ \text{rank}(X) &= n - \nu(X) \\ \text{rank}(X^\top) + \nu(X^\top) &= m \\ \text{rank}(X^\top) &= m - \nu(X^\top) \end{aligned}$$

Yet we know that $\text{rank}(X) = \text{rank}(X^\top)$. So we have:

$$\nu(X^\top) = \nu(X) + m - n.$$

(b) Similar to (a) we write out the rank equations for the matrices

$$\begin{aligned} \text{rank}(A) &= k - \nu(A) \\ \text{rank}(B) &= n - \nu(B) \\ \text{rank}(AB) &= n - \nu(AB). \end{aligned} \quad (3)$$

Substituting into $\text{rank}(A) + \text{rank}(B) - k \leq \text{rank}(AB)$ we get

$$\begin{aligned} \text{rank}(A) + \text{rank}(B) - k &\leq \text{rank}(AB) \\ (k - \nu(A)) + (n - \nu(B)) - k &\leq n - \nu(AB) \\ n - \nu(A) - \nu(B) &\leq n - \nu(AB) \\ \nu(A) + \nu(B) &\geq \nu(AB), \end{aligned}$$

where in that last step we both subtracted both sides by n and multiplied both sides of the inequality by -1 switching its direction. And we are done.

(c) Similarly to (b) let's consider equation (3). If we consider what $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$ is actually saying we have that we have both $\text{rank}(AB) \leq \text{rank}(A)$ and $\text{rank}(AB) \leq \text{rank}(B)$.

We see that

$$\begin{aligned} n - \nu(AB) \leq n - \nu(A) &\iff \nu(A) \leq \nu(AB), \\ n - \nu(AB) \leq n - \nu(B) &\iff \nu(B) \leq \nu(AB). \end{aligned}$$

So we have

$$\max\{\nu(A), \nu(B)\} \leq \nu(AB)$$

(d) We compute AB and BA for the sake of discussion

$$AB = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 \quad \text{and} \quad BA = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

We also look at what the null spaces of the respective matrices are. So we describe our linear transformations as functions

$$A = \begin{bmatrix} 0 & 1 \end{bmatrix} = (x, y) \mapsto y \quad \text{and} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \hat{x} \mapsto \begin{bmatrix} \hat{x} & \hat{0} \end{bmatrix},$$

where $\hat{x} = [x_1 \ \cdots \ x_n]$ and $\hat{0}$ is the n -ary zero vector. Given that $\ker(A) = \{(x, 0) : x \in \mathbb{F}\}$ we have that $\nu(A) = 1$. Also, notice $\ker(B) = \{\hat{0}\}$, so $\nu(B) = 0$. Lastly notice that both $\nu(AB) = 1$ by AB being the zero map, and $\nu(BA) = 1$ by single element linear transformation. But then we have $\max\{0, 1\} \leq 1$ in either case which does not help us. Hm.

So we consider the following counterexample

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

So,

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The nullity of A is 1 because our rank is 2. The nullity of B is 0, and finally the nullity of AB is 0 because it's invertible. So we truly are stuck with $\max\{0, 1\} \leq 0$ which doesn't hold. $\square$

P.4.13. Let $\mathcal{V}$ be a nonzero n -dimensional vector space over the field $\mathbb{F}$, and let $T \in \mathfrak{L}(\mathcal{V}, \mathbb{F})$ be a nonzero linear transformation. Why is $\dim(\ker(T)) = n - 1$?

Proof. Let $T : \rightarrow \mathbb{F}$ be a linear transformation. Now the $\text{rank}(T) \leq 1$, and because T is nonzero, we must have $\text{rank}(T) = 1$. By the rank nullity theorem we see that

$$\begin{aligned} 1 + \nu(T) &= n \\ \nu(T) &= n - 1. \end{aligned}$$

Done. □

P.4.35. For each of the following matrices, compute its reduced row echelon form, its pivot column decomposition, and bases for its column space and null space.

$$(a) \begin{bmatrix} -1 & -2 & 2 & 3 & 3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 1 & 0 & 1 & 1 & 1 \\ 3 & 2 & 2 & 3 & 2 \end{bmatrix}.$$

Solution. (a) We begin by performing Gauss-Jordan elimination on the matrix without augmentation.

$$\begin{bmatrix} -1 & -2 & 2 & 3 & 3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix}$$

$$r_1 : -r_1$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix}$$

$$r_2 : r_2 - 3r_1, \text{ and } r_3 : r_3 - 2r_1$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 7 & 14 & 8 \\ 0 & 0 & 3 & 6 & 8 \end{bmatrix}$$

$$r_2 : r_2/7$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 1 & 2 & \frac{8}{7} \\ 0 & 0 & 3 & 6 & 8 \end{bmatrix}$$

$$r_3 : r_3 - 3r_2$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 1 & 2 & \frac{8}{7} \\ 0 & 0 & 0 & 0 & \frac{32}{7} \end{bmatrix}$$

$$r_3 : 7/32r_3$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 1 & 2 & \frac{8}{7} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$r_1 : r_1 + 3r_3$, and $r_2 : r_2 - (8/7)r_3$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Finish with $r_1 : r_1 + 2r_2$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns indices are 1, 3, 5. We can write $2a_1 = a_2$ and $a_4 = a_1 + 2a_3$. We take our basis for $\text{col}(A)$ as the original columns in our pivot column indices as

$$\left\{ \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

For the null space, we need to solve $Ax = 0$. Letting $x = [x_i]$ with $i = 1, \dots, 5$, we expand to get

$$x_1 + 2x_2 + x_4 = 0$$

$$x_3 + 2x_4 = 0$$

$$x_5 = 0$$

For simplicity, we remove subscripts as $x_2 = y$ and $x_4 = z$ which gives us

$$x_1 = -2y - z, \quad x_3 = -2z, \quad x_5 = 0$$

This gives us a basis for our null space as:

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

(b) We begin by again using Gauss-Jordan elimination for our row-reduction algorithm. The computation follows:

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 1 & 0 & 1 & 1 & 1 \\ 3 & 2 & 2 & 3 & 2 \end{bmatrix}$$

$$r_2 : r_2 - r_1 \text{ and } r_3 : r_3 - 3r_1$$

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 0 & -2 \\ 0 & 5 & -4 & 0 & -7 \end{bmatrix}$$

$$r_3 : r_3 - 5r_2$$

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}$$

$$r_2 : r_2 + r_3$$

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}$$

Finally, we perform $r_1 : r_1 - 2r_3$, and $r_1 : r_1 + r_2$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}.$$

The pivot columns here are 1, 2, 3. So our basis from our original matrix follows as:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}.$$

To get a basis for the nullity of B we expand $Bx = 0$, giving us the following equations

$$x_1 + x_4 - 2x_5 = 0$$

$$x_2 + x_5 = 0x_3 + 3x_5 = 0$$

So $x_1 = -x_4 + 2x_5$, $x_2 = x_5$, and $x_3 = -3x_5$. Then we let x_4 and x_5 range freely to see that the basis for our null space is

$$\left\{ \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

□

P.4.38. Suppose that $A \in \mathbf{M}_{6 \times 4}$ is row equivalent to

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- (a) Find a basis for $\text{null}(A)$.
- (b) What is $\dim(\text{null}(A^\top))$

Solution.

□